

Catalog (or dictionary)

- ❑ It is the set of tables containing the information the system itself needs to work
- ❑ Its contents are metadata (data regarding data)
- ❑ Its structure and contents differ from one DBMS to another, but there are standardized views

“Static” contents of the catalog

- ❑ System parameters
 - Size of buffers pool
 - Page size
- ❑ Table information
 - Table name, file name and kind of file
 - Name and type of each attribute
 - Indexes defined in the table
 - Integrity constraints
- ❑ Index information
 - Name and structure
 - Indexed attributes
- ❑ View information
 - Name and definition
- ❑ User information
 - Permissions

“Dynamic” contents of the catalog

□ Tables

- Cardinality: Number of tuples
- Size: Number of blocks

□ Indexes

- Cardinality: Number of different values
- Size: Number of blocks
- Depth: Number of non-root levels
- Range: Minimum and maximum existing values

□ Users

- Connection attempts

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Alberto Abelló

3

SQL'03 catalog views (I)

□ DEFINITION_SCHEMA

- Contains 61 “tables”
- Used by the administrator

□ INFORMATION_SCHEMA

- Contains 71+43 views
 - Most of them coincide with DEFINITION_SCHEMA
- Used by users

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Alberto Abelló

4

SQL'03 catalog views (II)

- ▣ ASSERTIONS
- ▣ ATTRIBUTES
- ▣ CHECK_CONSTRAINTS
- ▣ COLUMNS
- ▣ REFERENTIAL_CONSTRAINTS
- ▣ ROUTINES
- ▣ SEQUENCES
- ▣ TABLE_PRIVILEGES
- ▣ TABLES
- ▣ TRIGGERS
- ▣ VIEWS
- ▣ ...